

BEE 208 — Question 18: Stop Load Test When Limit Is Reached

Control Statement: for loop with break | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables powerRating, totalLoad (double) and i (int)

Step 3: Set totalLoad = 0

Step 4: Use a for loop from i = 1 to 20

Step 5: Prompt user to enter power rating of appliance i

Step 6: Read powerRating

Step 7: totalLoad = totalLoad + powerRating

Step 8: If totalLoad > 5000, display warning and break

Step 9: Else display running total

Step 10: End loop

Step 11: Display final totalLoad

Step 12: End

2. PSEUDOCODE

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BEGIN

DECLARE powerRating, totalLoad AS DOUBLE

DECLARE i AS INTEGER

SET totalLoad = 0

FOR i = 1 TO 20 DO

    OUTPUT 'Appliance', i, 'Power Rating (W):'

    INPUT powerRating

    totalLoad = totalLoad + powerRating

    IF totalLoad > 5000 THEN

        OUTPUT 'Maximum load exceeded. Stop adding appliances.'

        BREAK

    ELSE

        OUTPUT 'Running Total Load:', totalLoad, 'W'

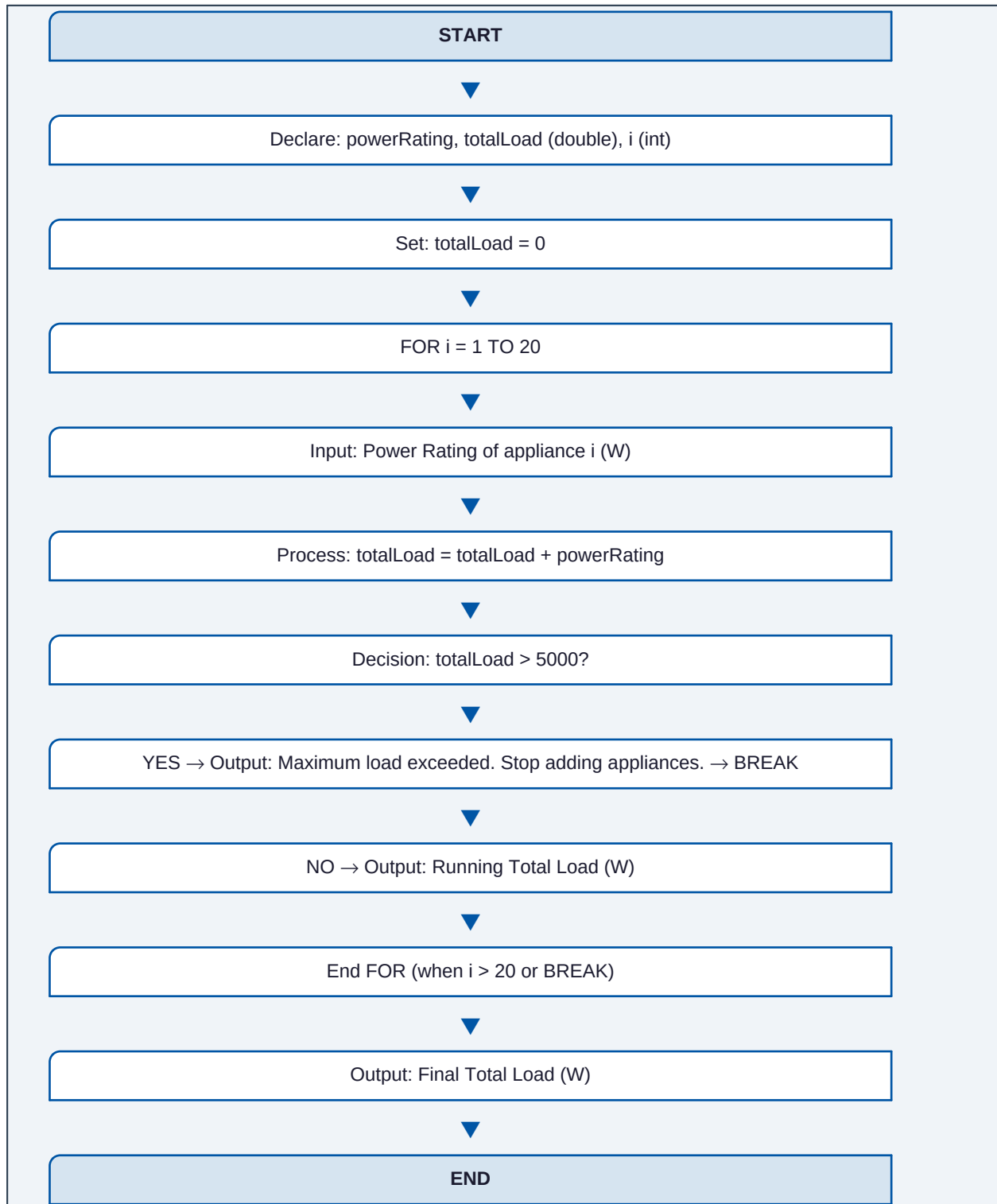
    END IF

END FOR

OUTPUT 'Final Total Load:', totalLoad, 'W'
```

END

3. FLOWCHART (Structured Diagram)



4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	Appliances: 1000W×5 = 5000W, then 200W	Expected Output	Warning at 5200W

Input	20 appliances each 100W	Expected Output	Total = 2000 W (no break)
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